Editors' Suggestion

# Transition from antiferromagnets to altermagnets: Symmetry-breaking theory

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(Received 18 August 2024; revised 9 June 2025; accepted 16 September 2025; published 14 October 2025)

Considering the similarity of the real-space configurations for the opposite spin sublattices in both antiferromagnets (AFMs) and altermagnets (AMs), the relationship between them should be profound. In this work, we demonstrate that AFMs and AMs can be connected with spin groups and their subgroups. Consequently, the breaking of the combined inversion or translation operation with time-reversal symmetry ($\mathcal{PT}$ or $\mathbf{t}\mathcal{T}$) in AFMs will induce transition from AFM to AM. We systematically list all collinear spin point groups and space groups that can realize the transition for the three types of AFMs: $\mathcal{PT}$, $\mathbf{t}\mathcal{T}$, and $\mathcal{PT}$-$\mathbf{t}\mathcal{T}$. Moreover, we propose that Floquet engineering using circularly polarized light and surface cutting engineering are effective approaches to break $\mathcal{PT}$ and $\mathbf{t}\mathcal{T}$ symmetries of AFM, respectively, achieving the transition. Interestingly, the features and magnitude of AM spin splitting can be tuned by adjusting various parameters of Floquet engineering. In our work, we not only establish a theoretical framework for the transition from AFM to AM but also provide practical approaches utilizing the achievements in AFMs for 100 years to obtain AMs, significantly expanding the scope of AM materials for both theoretical studies and future practical applications.

DOI: 10.1103/cxvd-mp2h

## I. INTRODUCTION

Recently, altermagnets (AMs) have drawn significant interest from the scientific community following their theoretical proposal and experimental confirmation [1–19]. In this magnetic phase, local magnetic moments in real space exhibit antiferromagnetic (AFM) coupling akin to that of AFMs, whereas bands demonstrate alternate spin polarization in reciprocal space. Unlike conventional AFMs whose opposite spin sublattices are connected through translation or inversion symmetries, in AMs, the opposite spin sublattices are linked by crystal-rotation symmetries, potentially augmented by additional translation or inversion operations [1]. Although the different symmetry requirements result in distinct spin groups to describe AFM and AM systems, considering the similarity of real-space configuration of the opposite spin sublattices in AFMs and AMs, the two types of spin groups may be connected by group-subgroup relationship. Namely, the spin group of AMs is one of the subgroups of AFMs. If the connection is to be built, we can obtain an AM from an AFM through symmetry breaking. Such a transition will not only significantly expand the scope of AM materials based on the achievements in AFMs for 100 years but also provide an effective modulation approach for AFM spintronics [20,21]. However, there is still a lack of systematic exploration of the symmetry-connection and symmetry-breaking methods between the two magnetic phases.

Until now, several primary approaches for converting AFMs into AMs have been proposed that can be categorized as van der Waals (vdW) stacking engineering of two-dimensional (2D) materials [22–24], Janus structures [25,26], and external electric fields [27–30]. The vdW stacking engineering involves weak magnetic exchange coupling and subsequently low Néel temperature. Janus structures will introduce an extra disorder-altered intrinsic structure of materials. Although electric field modulation offers a noninvasive method to break $\mathcal{PT}$ symmetry, making it particularly promising for spintronic applications [27,31,32], it is limited with constraints of the directionality and symmetry-breaking types. All the above methods are primarily applicable to 2D materials. However, experimentally realized 2D AFM materials remain relatively scarce, whereas three-dimensional (3D) AFM materials are quite abundant, such as tetragonal CuMnAs [33], cubic NiO [34], and *G*-type $Cr_2O_3$ [35], among others. Furthermore, a general understanding and approach for the transition from AFM to AM are still lacking. Therefore, developing a theoretical framework of the transition from AFM to AM and effective methods to induce the magnetic transitions in both 2D and 3D collinear AFMs are essential for the research advancement in this field.

In this work, we systematically analyze the symmetry relationships between collinear AFMs and AMs through group-subgroup relationships and identify all possible magnetic transition pathways driven by symmetry breaking. For $\mathcal{PT}$-symmetry related AFMs ($\mathcal{PT}$-type AFMs), we identify 18 point group-subgroup pairs corresponding to 82 transition pathways, encompassing 252 spin space groups describing AFM systems to 422 spin space groups describing AM systems. There are 155 and 163 spin space groups for $\mathbf{t}\mathcal{T}$-symmetry connected AFMs ($\mathbf{t}\mathcal{T}$-type AFMs) and

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AFMs possessing both $\mathcal{PT}$ and $\mathbf{t}\mathcal{T}$ symmetries (classified as $\mathcal{PT}$-$\mathbf{t}\mathcal{T}$-type AFMs), respectively, which can realize such a transition. We propose two practical methods: Floquet engineering and surface engineering to induce such transitions via $\mathcal{PT}$- and $\mathbf{t}\mathcal{T}$-symmetry breaking. The magnetic transitions by breaking $\mathcal{PT}$ and/or $\mathbf{t}\mathcal{T}$ symmetry are thoroughly discussed with tight-binding models and confirmed with first-principles calculations with typical real AFMs $Cr_2O_3$, $CaFeO_2$, KMnP, and NiO.

## II. SYMMETRY ANALYSIS

Here, we focus on the transition between collinear AFMs and AMs due to the collinear nature of AMs [1,2,36]. In conventional collinear AFM materials, opposite spin sublattices are connected by either space inversion or/and translation with time-reversal symmetry, namely, AFMs can be categorized by $\mathcal{PT}$-type AFMs, $\mathbf{t}\mathcal{T}$-type AFMs, and $\mathcal{PT}$-$\mathbf{t}\mathcal{T}$-type AFMs. However, in AM materials, the opposite spin sublattices are connected by symmetry operations other than space inversion or pure translation (we name the operation as $AT$). Therefore, the key to the transition from AFMs to AMs lies in breaking space inversion or/and translation symmetry while preserving other symmetries to connect opposite spin sublattices. It is important to note that, in collinear AFMs and AMs, spin orientations are limited to two antiparallel directions and remain unchanged across magnetic transitions. Thus, time-reversal and spin-rotation symmetries are preserved, while certain combined symmetries with crystalline operations may be broken.

We first study the symmetry evolution of the $\mathcal{PT}$-type AFM with the point group symmetry of the crystal. When spin freedom is disregarded, crystal symmetries are described by 32 point groups. The group-subgroup relationship among these 32 point groups is shown in Fig. 1, where pink and arctic blue boxes represent point groups with and without space inversion symmetry, respectively. As shown in Fig. 1, we have identified 18 potential pathways (indicated by green lines) that break space inversion symmetry, through which the transition from AFMs to AMs can be realized.

The point groups alone are insufficient to fully describe the symmetries of the two types of collinear magnetic materials. It is necessary to consider the spin point groups and spin space groups [37–39]. For the two types of magnetic states, we must use spin group $\mathbf{r}_s \times \mathcal{R}_s^{III}$, where $\mathbf{r}_s$ is the spin-only group, and $\mathcal{R}_s^{III}$ is the third type of nontrivial spin groups [1,2]

$$\mathcal{R}_s^{III} = [E\|\mathcal{N}] \oplus [C_2\|\mathcal{G} - \mathcal{N}], \tag{1}$$

where $\mathcal{N}$ is a halving subgroup of a crystallographic point group $\mathcal{G}$, and it exclusively encompasses real-space transformations that interchange atoms within the same-spin sublattices, whereas the coset $\mathcal{G} - \mathcal{N}$ includes only real-space transformations that interchange atoms between opposite-spin sublattices, which can be written as $\mathcal{A}\mathcal{N}$, where $\mathcal{A}$ denotes any operations among them. According to the labels of spin-group symmetries [1,37–39], the operations in $\mathcal{N}$ and $\mathcal{A}\mathcal{N}$, respectively, can be written as $[E\|A]$ and $[C_2\|B]$, where the $E$ and $C_2$ in front of the double vertical line represent spin-space identity and inversion, respectively. Here, $A$ and $B$ can be any real-space point group operations. The labels of $[C_2\|B]$, $[C_2\|P]$, and $[C_2\|\mathbf{t}]$, respectively, represent the conventional $\mathcal{PT}$ or $\mathbf{t}\mathcal{T}$ operation. It is worth emphasizing that, in the absence of spin-orbit coupling (SOC), type-III magnetic point groups [40] and collinear nontrivial spin groups are built in a similar manner, and both have been employed to describe AFM or AM materials. The process of their construction involves selecting a crystalline point group, then identifying a subgroup of index 2. The operations of this subgroup are used to link the magnetic sublattices with the same spin direction, while the remaining half of the operations connect sublattices with opposite spins. By examining all 32 crystalline point groups, we find that 58 type-III magnetic point groups or 58 nontrivial spin groups can be derived.

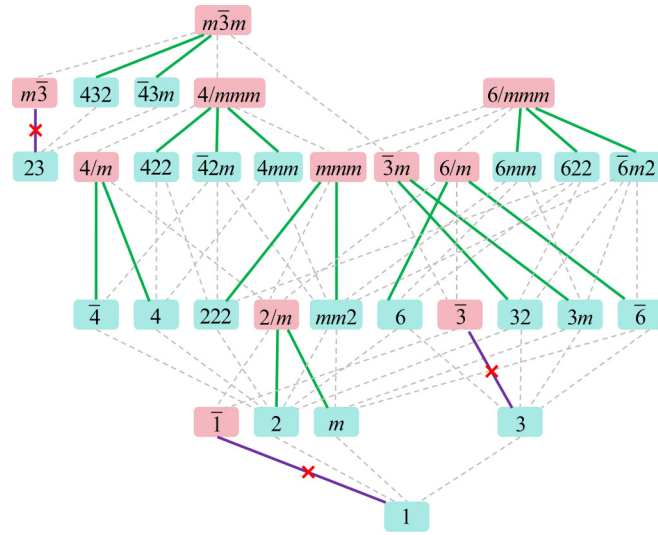



FIG. 1. The evolutionary relationship between point groups and their subgroups. The point groups with inversion symmetry are denoted by pink boxes, while those without inversion symmetry are represented by arctic blue boxes. Transition pathways from groups with inversion symmetry to their subgroups without it are indicated by both green and purple lines. However, only the green lines may describe the transition from AFM to AM phases.

We first discuss the case of breaking $[C_2\|P]$ symmetry in nontrivial spin groups [1,37–39]. For a spin point group that used to describe AFM materials, the space inversion operation $P$ must be included in $\mathcal{A}\mathcal{N}$. However, for AM materials, $P$ cannot be included in $\mathcal{A}\mathcal{N}$; instead, at least one rotation or rotation-inversion combined operation must be presented in $\mathcal{A}\mathcal{N}$. Therefore, if $\mathcal{A}\mathcal{N}$ for a spin point group contains both $P$ and other rotation or rotation-inversion operations, breaking the inversion operation $P$ while preserving the rotation or rotation-inversion operations should guarantee a transition from AFMs to AMs. If we disregard the transformation of spin freedom, it is sufficient to analyze the relationship between different point groups. As shown in Fig. 1, there are 21 such possible pathways to break the inversion operation in the point group. However, three pathways denoted by purple lines will not preserve any operations in the coset. Therefore, there are only 18 potential pathways denoted by the green lines in Fig. 1 for the transition from AFMs to AMs. The 18 possible pathways achieving the space inversion symmetry breaking correspond to nine

original crystalline point groups, which correspond to 19 collinear spin point groups possessing inversion symmetry $P$ in the coset $\mathcal{A}\mathcal{N}$, as shown in Figs. S1–S10 in the Supplemental Material [41]. By analyzing all their subgroups that break inversion symmetry, we identified 82 potential pathways for the magnetic transitions from AFM to AM phase, as consolidated in Table S2 in the Supplemental Material [41]. It is worth noting that, if the inversion operation is included in $\mathcal{N}$ in an AM material, it cannot be derived from the corresponding AFM one. Like the $\mathcal{PT}$-symmetric counterpart, $\mathbf{t}\mathcal{T}$-type and $\mathcal{PT}$-$\mathbf{t}\mathcal{T}$-type AFMs can undergo a magnetic transition to AMs through the breaking of $\mathbf{t}\mathcal{T}$ or/and $\mathcal{PT}$ symmetries. Our in-depth analysis of the group-subgroup relationship shows that the $\mathbf{t}\mathcal{T}$-type AFM transition can be realized by 155 spin space groups, while the $\mathcal{PT}$-$\mathbf{t}\mathcal{T}$-type AFM transition can be realized by 163 spin space groups. Further details are available in Sec. I in the Supplemental Material [41].

## III. FLOQUET ENGINEERING

We then investigate general and feasible methods breaking $\mathcal{PT}$ or/and $\mathbf{t}\mathcal{T}$ symmetries to induce the magnetic transition as mentioned above. Regarding the $\mathcal{PT}$ symmetry breaking, applying a vertical electric field is an effective method to induce the magnetic transition, as discussed in detail for AFM MoTe in Sec. II in the Supplemental Material [41]. However, this approach is limited to 2D materials. Here, we propose Floquet engineering is a general and feasible approach suitable for both 2D and 3D AFM materials to realize such a transition. Floquet engineering is a sophisticated technique in quantum physics and condensed matter systems, which utilizes time-periodic perturbations to dynamically control and manipulate the properties of materials. Authors of recent studies have highlighted its ability to control the electronic, magnetic, and topological properties of materials [42–44]. In this work, we find that applying circularly polarized light is a general approach to break $\mathcal{PT}$ symmetry in AFMs, facilitating their magnetic transition to AMs.

To demonstrate the symmetry-breaking mechanism of Floquet engineering, we use the AFM honeycomb structure as an example, as shown in Fig. 2(a). In the absence of an external field, its Hamiltonian can be expressed as

$$\mathcal{H} = t \sum_{\langle i,j\rangle,\sigma} c^{\dagger}_{i,\sigma} c_{j,\sigma} - J_{sd} \sum_{i,\sigma,\sigma'} S_i \cdot c^{\dagger}_{i,\sigma} \sigma_{\sigma\sigma'} c_{i,\sigma'} + \text{H.c.}, \quad (2)$$

where the term $c^{(\dagger)}_{i,\sigma}$ represents the annihilation (creation) operator for an electron at site $i$ with spin $\sigma$. The nearest-neighbor hopping strength is denoted by $t$, and the coupling constant $J_{sd}$ characterizes the on-site exchange interaction between the spins of the itinerant electrons and the localized spins at the magnetic sites $S_i$. The AFM honeycomb lattice is described by the spin point group ${}^{2}6/{}^{1}m^{2}m^{1}m$. Due to the protection of $\mathcal{PT}$ symmetry, the spin-up and spin-down bands remain degenerate throughout the whole Brillouin zone (BZ), as shown in Fig. 2(b). When left circularly polarized light, described by $\mathbf{A}(t) = A[\cos(\omega t), -\sin(\omega t), 0]$ (with photon energy $\hbar\omega = 10\,\text{eV}$) is applied to the system, the hopping interactions are modified by an additional phase factor

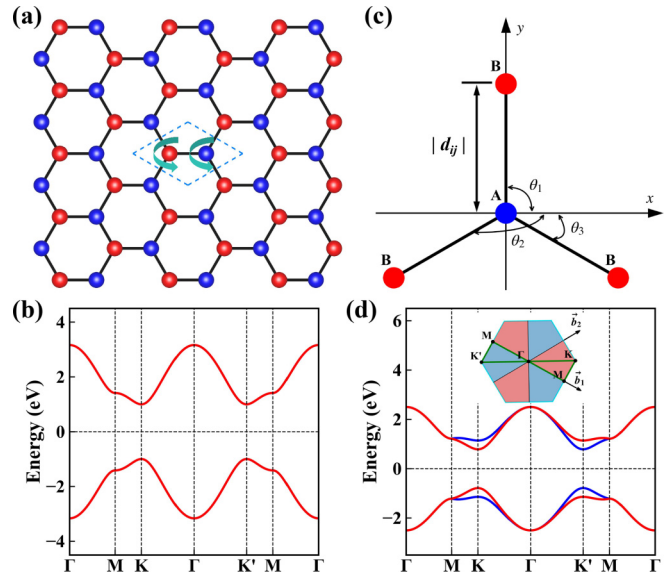



FIG. 2. AFMs to AMs by Floquet engineering in $\mathcal{PT}$ AFM honeycomb lattice. (a) AFM honeycomb lattice. Blue and red represent the spin-up and spin-down sublattices, respectively. (b) The spin-degenerate energy bands in the AFM honeycomb lattice. (c) Floquet hopping of the honeycomb lattice. (d) The energy band structure with left circularly polarized light intensity 1.0/Å. The inset illustration shows 2D BZ with AM spin splitting. Blue and red represent spin-up and spin-down, respectively.

$\exp[i\frac{e}{\hbar}\mathbf{A}(t)\cdot\mathbf{d}_{ij}]$, where $\mathbf{d}_{ij}$ is the relative position vector between orbitals $i$ and $j$. According to Floquet-Bloch theory under the high-frequency approximation [45–48], this time-dependent modulation can be effectively replaced by a time-independent term of the form $\exp(-iq\theta)i^{q}J_q(\frac{e}{\hbar}A)$, where $\theta$ is the angle between $\mathbf{d}_{ij}$ and the $x$ axis [see Fig. 2(c)], and $J_q$ denotes the Bessel function of the first kind of order $q$ (a more detailed derivation can be found in the Supplemental Material [41]). As a result, the effective Floquet hopping modulated by the light field breaks the $\mathcal{PT}$ symmetry of the system. This process can also be viewed as applying a pseudomagnetic vector of the same direction to the opposite spin sublattice, as shown in Fig. 2(a). Detailed symmetry analysis reveals that the system breaks $\mathcal{PT}$ and $[C_2\|C_{6z}]$ symmetries and maintains $[E\|C_{3z}]$ and $[E\|M_z]$ symmetries, i.e., the spin point group changes to ${}^{1}\bar{6}^{2}m^{2}2$. As shown in Fig. 2(d), the energy bands exhibit the typical characteristics of AM spin splitting, with opposite spin sublattices connected by three in-plane $C_2$ or three vertical mirror symmetries. It is important to note that, if the left circularly polarized light is replaced with right circularly polarized light, the spin splitting can be reversed, as illustrated in Fig. S20(b) in the Supplemental Material [41]. Additionally, the magnitude of spin splitting can be adjusted by altering the light intensity. For example, the spin splitting at the $K$ point can increase from 0.356 to 0.454 eV when the value of $A$ changes from $1.0/\text{Å}^{-1}$ to $1.2/\text{Å}^{-1}$, as shown in Figs. S20(a) and S20(d) in the Supplemental Material [41]. These findings suggest that, through Floquet engineering, we can achieve spin-tunable AM materials. Moreover, if we remove the light field, the system would restore the equilibrium AFM state. This dynamic control provides an ideal platform

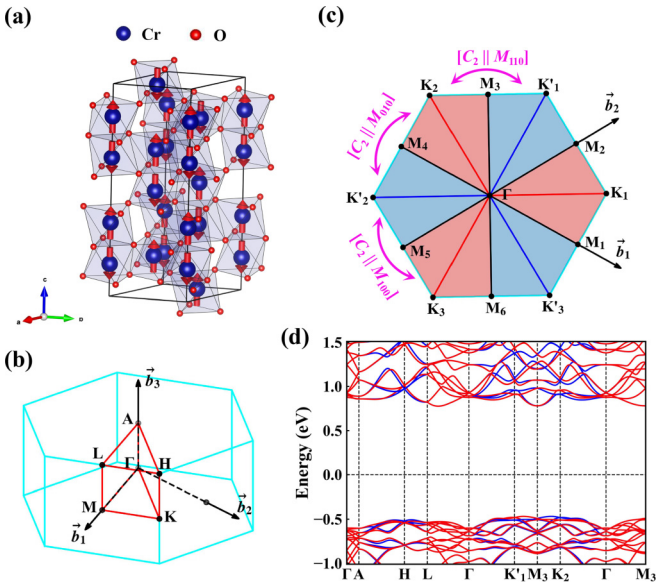



FIG. 3. AFMs to AMs by Floquet engineering in $G$-type AFM $Cr_2O_3$. (a) and (b) Crystal structure and the 3D BZ of $G$-type AFM $Cr_2O_3$, respectively. (c) The BZ plane at $k_z = 0$ exhibits AM spin splitting. (d) Energy bands under left circularly polarized light ($\hbar\omega$ = 15 eV) with an intensity of 0.35/Å, where blue and red denote spin-up and spin-down states, respectively.

for the ultrafast and reversible generation of spin currents. Therefore, our general theory and practical methods not only expand the class of AM materials to include traditional high Néel temperature AFMs but also pave the way for efficient and ultrafast control of spintronic functionalities.

To confirm the symmetry breaking of Floquet engineering on $\mathcal{PT}$-type AFMs, we take $G$-type AFM $Cr_2O_3$ as a typical example. Here, $Cr_2O_3$ is notable for its exceptionally high AFM ordering temperature of ~300 K [49], and it has drawn considerable attention for its intriguing magnetoelectric properties [50]. As depicted in Fig. 3(a), the adjacent magnetic Cr atoms in each vertical row show opposite spins, exhibiting the characteristics of $G$-type AFM order. The symmetries of the material can be described by the spin point group ${}^2\bar{3}^2m$. The energy bands of $Cr_2O_3$ remain spin degenerate across the entire BZ, as shown in Fig. S22(b) in the Supplemental Material [41], due to the presence of $\mathcal{PT}$ symmetry. When left circularly polarized light is applied along the $z$ direction, the $\mathcal{PT}$ symmetry of the system is broken, leading to AM spin splitting, as illustrated in Figs. 3(c) and 3(d). Furthermore, the light field induces pseudomagnetic vectors, disrupting the in-plane $C_2$ symmetry. The spin point group of $Cr_2O_3$ now changes to be ${}^13^2m$. Consequently, the opposite spin sublattices of $Cr_2O_3$ are now connected by three vertical mirror symmetries: $[C_2\|M_{100}]$, $[C_2\|M_{010}]$, and $[C_2\|M_{110}]$, as shown in Fig. 3(c). The three vertical mirror symmetries will protect the formation of three vertical spin-degenerate nodal planes within the 3D BZ. It is worth mentioning that $Cr_2O_3$ is a typical 3D material. For 2D materials, Floquet engineering is also suitable for inducing the transition from AFM to AM through $\mathcal{PT}$ symmetry breaking. A typical example of 2D MnSr can be found in Sec. III in the Supplemental Material [41].

## IV. SURFACE ENGINEERING

In addition to $\mathcal{PT}$-type AFMs, $\mathbf{t}\mathcal{T}$-type AFMs are also a large class of traditional AFMs ($\mathbf{t}$ is a fractional translation along the lattice basis vector direction, and here, we assume without $\mathcal{PT}$ symmetry in the $\mathbf{t}\mathcal{T}$-type AFM connecting the opposite spin sublattices). If SOC is ignored, their symmetries must be described by spin space groups. For this type of AFM, we propose that surface engineering can convert them into AMs. After slicing a surface to intercept the fractional translation of the $\mathbf{t}\mathcal{T}$-type AFM, an interface is formed with the vacuum along the Miller plane ($hkl$). We define $\hat{\mathbf{n}}$ as the unit vector in the normal direction of the surface, and it is a key parameter to characterize the surface symmetry. As discussed by Weber *et al.* [51], the surface symmetry group is a subgroup of the group of the bulk. Here, we label it as $\mathbf{G}_{\hat{\mathbf{n}}} = [\mathbf{R}_i\|(\mathbf{R}_j^{\hat{\mathbf{n}}}|\mathbf{t}^{\perp\hat{\mathbf{n}}})]$, as shown in Fig. S23 in the Supplemental Material [41]. Because we concentrate on the surface that hosts AMs, we only study the surface hosting equal opposite spin sublattices that compensate each other. Such surfaces are different from the surface with equilibrium magnetization in previous reports [51,52]. Provided no $\mathbf{t}\mathcal{T}$ operations connect the opposite spin sublattices on the surface, while other compound operations link the opposite spin sublattices, the AM will emerge. By analyzing all 289 spin space groups of $\mathbf{t}\mathcal{T}$-type AFMs, we find 155 among them host AM-like subgroups, namely, the systems with these spin space groups can be potentially turned into AMs with $\mathbf{t}\mathcal{T}$ symmetry breaking.

To illustrate the transition driven by $\mathbf{t}\mathcal{T}$ symmetry breaking in the $\mathbf{t}\mathcal{T}$-type AFM class, we consider the pure $\mathbf{t}\mathcal{T}$ AFM $CaFeO_2$. This compound crystallizes in a layered structure with a high-spin, distorted square-planar configuration and exhibits $G$-type AFM ordering with a Néel temperature of 420 K [53]. The magnetic moments of the Fe atoms exhibit AFM coupling, forming an in-plane magnetic order within the (001) plane. Additionally, antiparallel alignment of Fe spins along the $c$ axis leads to a doubling of the lattice constant $c$ in the magnetic unit cell, as depicted in Fig. S24(a) in the Supplemental Material [41]. In this compound, the two spin sublattices are related by the combined symmetry operation $\mathbf{t}\mathcal{T}$, where $\mathbf{t} = (0, 0, \frac{1}{2})$. The corresponding spin space group is given by $G_{ss} = Pc^1\bar{4}^12_1{}^1c^{\infty m}1$. The electronic band structure of $CaFeO_2$ displays semiconducting behavior with a band gap of 1.80 eV. Due to the presence of $\mathbf{t}\mathcal{T}$ symmetry, the band structure is entirely spin degenerate throughout the BZ, as shown in Fig. S24(b) in the Supplemental Material [41]. To break $\mathbf{t}\mathcal{T}$ symmetry, we construct a slab geometry with its surface normal oriented along the $\mathbf{t}$ direction. This breaks $\mathbf{t}\mathcal{T}$ symmetry, resulting in a change in the spin space group to $P^1\bar{4}^{-1}2_1{}^{-1}m^{\infty m}1$. In this group, the symmetries $[C_2\|M_{110}]$ and $[C_2\|M_{1\bar{1}0}]$ are preserved. As a consequence, spin degeneracy is lifted except along the $\tilde{\Gamma}$-$\tilde{M}$ high-symmetry line of 2D surface BZ, as shown in Fig. S24(d) in the Supplemental Material [41]. This leads to a characteristic $d$-wave AM spin distribution, illustrated in Fig. S24(c) in the Supplemental Material [41].

In addition to $\mathcal{PT}$- and $\mathbf{t}\mathcal{T}$-type AFMs, there are AFMs whose opposite spin sublattices are simultaneously connected by both $\mathcal{PT}$ and $\mathbf{t}\mathcal{T}$ symmetries, named as $\mathcal{PT}$-$\mathbf{t}\mathcal{T}$-type AFMs. To induce the transition from $\mathcal{PT}$-$\mathbf{t}\mathcal{T}$-type AFM to

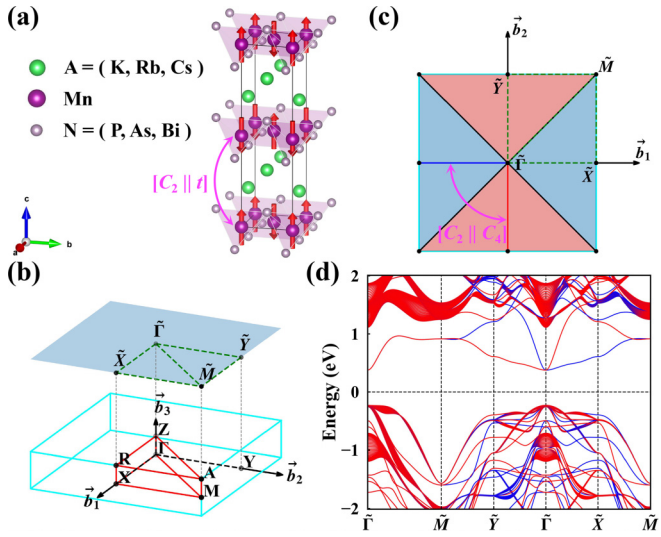



FIG. 4. Surface engineering-induced transition from AFMs to AMs in $\mathbf{t}\mathcal{T}$ AFM materials $A\mathrm{Mn}N$ ($A$ = K, Rb, Cs; $N$ = P, As, Bi). (a) Crystal structure and (b) 3D BZ of $A\mathrm{Mn}N$. (c) Surface 2D BZ and AM spin splitting. (d) Slab band structure of KMnP with 15 layers along the $z$ direction, where blue and red represent spin-up and spin-down states, respectively.

AM, both $\mathbf{t}\mathcal{T}$ and $\mathcal{P}\mathcal{T}$ symmetries must be simultaneously broken. There are a total of 228 spin space groups of this type of AFM, among which 163 have AM-like subgroups that can potentially turn into AMs. Here, we use a class of alkali metal manganese pnictides $\mathcal{P}\mathcal{T}$-$\mathbf{t}\mathcal{T}$-type AFM $A\mathrm{Mn}N$ [54] ($A$ = K, Rb, Cs; $N$ = P, As, Bi) to illustrate the phase transition through surface engineering. The magnetic moments of Mn atoms in this type of compound exhibit AFM coupling in the $x$-$y$ plane. Moreover, Mn atoms are also antiparallel in the $c$-axis direction, leading to a doubling of the lattice constant $c$ of the magnetic unit cell, as shown in Fig. 4(a). In the materials, the opposite spin sublattices are connected by $\mathbf{t}T$ [$\mathbf{t} = (0, 0, \frac{1}{2})$] and $\mathcal{P}\mathcal{T}$. The spin space group of the system is $G_{ss} = P_c^{\bar{1}}4_2/{}^1n^1m^1c\,{}^{\infty m}1$. We take KMnP as a specific example, whose electronic band structure exhibits the characteristics of a semiconductor with a band gap of 1.34 eV. Due to the coexistence of $\mathcal{P}\mathcal{T}$ and $\mathbf{t}\mathcal{T}$ symmetries, its band structure is completely spin degenerate in reciprocal space, as shown in Fig. S22(d) in the Supplemental Material [41]. To break its $\mathbf{t}\mathcal{T}$ and $\mathcal{P}\mathcal{T}$ symmetries, we construct a slab structure with its surface oriented perpendicular to the $\mathbf{t}$ direction. The spin space group of the material changes to $P^{-1}4/{}^1n^{-1}m^1m\,{}^{\infty m}1$. In this spin space group, both $\mathbf{t}\mathcal{T}$ and $\mathcal{P}\mathcal{T}$ are broken, while $[C_2||C_4]$ symmetry is preserved, leading to spin splitting, except along the $\tilde{\Gamma}$-$\tilde{M}$ high-symmetry line in the 2D surface BZ. This results in $d$-wave AM spin distribution, as illustrated in Fig. 4(c).

To illustrate another type of transition from $\mathcal{P}\mathcal{T}$-$\mathbf{t}\mathcal{T}$-type AFM to AM, we take the AFM NiO [34] as an example. In the paramagnetic phase, the symmetry of rocksalt NiO belongs to cubic space group $Fm\bar{3}m$. While in the AFM phase, the local magnetic moment of Ni atoms aligns along the $[11\bar{2}]$ direction [55], and the symmetry of its magnetic primitive cell belongs to spin space group $C_c^12/{}^1c\,{}^{\infty m}1$. The crystal structure, BZ, and high-symmetry $k$ paths of monoclinic NiO can be found in Figs. S25(a) and S25(b) in the Supplemental Material [41]. The $\mathcal{P}\mathcal{T}$ and $\mathbf{t}\mathcal{T}$ symmetries force the energy bands to be spin degenerate throughout the BZ, as shown in Fig. S25(c) in the Supplemental Material [41]. To break $\mathbf{t}\mathcal{T}$ symmetry, we create a 2D AFM thin film of NiO through surface engineering, as depicted in Fig. S25(a) in the Supplemental Material [41]. The spin space group of 2D AFM NiO is $C^{-1}2/{}^1m\,{}^{\infty m}1$ that still retains $\mathcal{P}\mathcal{T}$ symmetry, resulting in spin-degenerate electronic energy bands, as shown in Fig. S25(e) in the Supplemental Material [41]. When we further apply left circularly polarized light along the [010] direction, the spin space group of 2D NiO transforms into $C^{-1}2\,{}^{\infty m}1$. In this group, the opposite spin sublattices are only connected by $[C_2\|C_{2_{010}}]$, leading to AM spin splitting. As illustrated in Figs. S25(d) and S25(f) in the Supplemental Material [41], the electronic bands of 2D NiO exhibit typical AM spin splitting under the influence of the light field.

## V. DISCUSSIONS

In summary, we develop a general symmetry-breaking theoretical framework and a universal implementation strategy via Floquet and surface engineering for the transition from AFM to AM. The theory and approaches offer a versatile and highly controllable platform for broad implications for both fundamental research and spintronic applications. First, the theoretical framework for the phase transition from AFM to AM provides practical approaches utilizing the achievements in AFMs for 100 years to obtain AMs, significantly expanding the scope of AM materials for both theoretical and experimental studies. Secondly, both Floquet and surface engineering provide a broad range of tunable abilities of the obtained AM systems, including polarization, photon energy, light amplitude, and incident orientations of periodic light fields and symmetry-dependent effects via crystallographic orientation of surface engineering. Finally, the periodic light fields not only provide a general $\mathcal{P}\mathcal{T}$ symmetry-breaking method but also contribute a feasible control scheme for ultrafast spin dynamics through switching the circular polarization of the light. Therefore, our general theoretical framework and practical modulation methods not only expand the class of AM materials to include traditional high Néel temperature AFMs but also pave the way for efficient and ultrafast control of spintronic functionalities.

Our framework goes beyond prior studies in which authors mainly stressed the distinction between the symmetry descriptions of AFMs and AMs. Instead, we articulate the underlying connection between the two: through suitable symmetry-breaking strategies, a transition from AFM to AM can be systematically realized. To this end, we propose two universal and experimentally relevant implementation routes—Floquet engineering and surface engineering—that enable AFM-to-AM phase transitions, with the notable advantage of being applicable to 3D materials. Importantly, our symmetry classification encompasses all three fundamental types of AFMs ($\mathcal{P}\mathcal{T}$, $\mathbf{t}\mathcal{T}$, and $\mathcal{P}\mathcal{T}$-$\mathbf{t}\mathcal{T}$). This unified theoretical framework, together with practical modulation strategies, not only extends the family of AM materials by incorporating conventional high Néel temperature AFMs but also provides a foundation

for efficient and ultrafast control of AFM spintronics. Crucially, the transition to the AM phase offers tunable physical responses inaccessible in the parent AFM state, such as the anomalous Hall effect, spin current generation, and nonlinear transport, thereby pointing toward concrete experimental realizations of physics beyond symmetry cataloging.

To realize optic-field-induced AM Floquet-Bloch states in AFM materials, a sufficiently strong light-matter interaction is critical to drive the formation of these states. Recent experimental evidence has demonstrated that increasing the laser intensity effectively enhances the light-matter coupling strength, with typical electric field strengths ranging from 0 to 1 V/Å. However, excessively high intensities may induce heating or nonlinear absorption, potentially damaging the intrinsic properties of the material. To mitigate this, optical cavities can be employed to significantly amplify the effective Floquet field at extremely low fluences, as demonstrated in recent studies [56]. Furthermore, the interaction between light and the material must account for the optical penetration depth. Given that recent advancements in Floquet engineering primarily focus on layered materials with finite thickness, selecting AFM materials of this type is crucial for observing the transition from AFM to AM states under optical driving. The material thickness should be carefully optimized—neither too thin nor too thick—with experimental results suggesting an optimal range of 5–20 nm to balance light penetration and surface effects, thereby enhancing the observability of Floquet-induced states.

## ACKNOWLEDGMENTS

This work is supported by the National Natural Science Foundation of China (Grants No. 12574070 and No. 1250042757), the Postgraduate Scientific Research Innovation Project of Hunan Province (Grant No. CX20240616), the China Postdoctoral Science Foundation (Grants No. 2025M773383 and No. GZC20252231), the China Postdoctoral Science Foundation-Hunan Joint Support Program (Grant No. 2025T002HN), the Excellent Youth Funding of Hunan Provincial Education Department (Grant No. 22B0175), and Natural Science Foundation of Hunan Province (Grant No. 2023JJ30572).

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